

POLICY DESIGN



IDEAS FOR TODAY

Previously:

Leaders need to think about their own incentives

Leaders need to provide the right incentives to those tasked with implementation of their ideas

Today:

Leaders need to understand how their policies create incentives

EXERCISE: REGULATING MPG

Form groups of 3-4

For this exercise, **ignore the existence of EV vehicles**

Problem: Transportation accounts for about 28–30% of U.S. greenhouse gas emissions

Main goal: Increase fuel economy of vehicles on roads

Roughly speaking, there are two categories of vehicles

- ▶ Cars: average 27 miles per gallon (MPG)
- ▶ Light trucks: average 23 MPG

Your regulatory tool is a lower bound on average MPG of vehicles sold by manufacturers in each category

EXERCISE: REGULATING MPG

A concern to consider:

- ▶ Light trucks are frequently work vehicles (farmers, construction, small businesses)
- ▶ Tightening MPG restrictions on those may increase the price and force drastic performance loss
 - ▶ This may be economically distortionary and politically hard

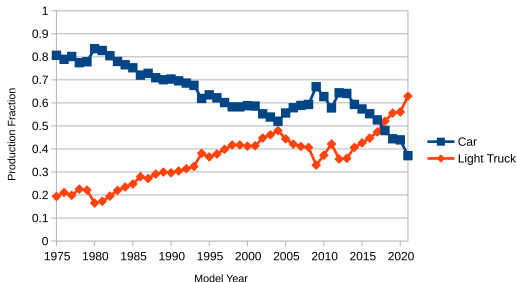
Which policy would you recommend and why:

1. Substantial increase in required MPG for both categories
2. Substantial increase in required MPG for cars only
3. No policy change

CORPORATE AVERAGE FUEL ECONOMY (CAFE)

CAFE standards were first enacted in 1975

Set separate standards for passenger cars vs. light trucks



Consumers shifted from cars to light trucks

More light trucks on roads may have increased fatalities

TAKE AWAYS FROM THE EXAMPLE

Everyone responds to incentives:

Leaders respond to incentives provided by selection rules

Workers respond to employment contracts

Citizens respond to incentives provided by policies

Three considerations for policy making:

1. Leaders must understand what incentives policies create
2. There are always trade-offs
3. Complex problems require complex solutions, but complex solutions create unintended behavior

ANTICIPATING BEHAVIOR CHANGES

Let U be the leader's payoff from a policy

The payoff depends on policy p and citizen's actions a :

$$U(p, a)$$

But citizen's strategically react to the policy:

$$a^*(p)$$

Strategic adjustment: response to policy incentives $a^*(p)$

When choosing p , the leader needs to anticipate $a^*(p)$ to understand the final payoff from the policy: $U(p, a^*(p))$

CAFE EXAMPLE

Fuel efficiency U is affected by policy, denoted by p

Citizens' purchases $a^*(p)$ are affected by policy

Let the pre-CAFE policy be p_0

Fixing the current purchasing behavior $a^*(p_0)$, CAFE standards would increase efficiency

$$U(p_1, a^*(p_0)) > U(p_0, a^*(p_0))$$

But drivers change purchasing behavior to $a^*(p_1)$

$$U(p_1, a^*(p_1)) < U(p_1, a^*(p_0))$$

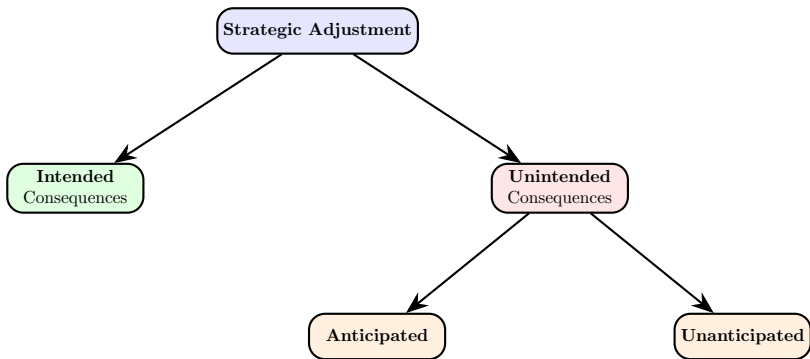
Policy effects may be smaller than intended or even negative

THIS IS THE KEY IDEA FROM GAME THEORY!

Think ahead: What will others do once you implement a policy?

Actions “up the tree” affect actions “down the tree”

TYPES OF STRATEGIC ADJUSTMENT



INTENDED STRATEGIC ADJUSTMENT

Intended strategic adjustment

- ⇒ Adjustment anticipated and welcome by the policy maker
- ⇒ Frequently the main point of the policy
- ⇒ Change in $a^*(p)$ that increases U

Example: Introduction of co-pay for medical services

- ▶ Meant to decrease government's cost of these services—direct effect even if people do not change behavior
- ▶ But is intended mainly to lower patients' unnecessary utilization of those services—through $a^*(p)$

Understanding strategic adjustment allows you to use a tool

- ▶ Intended for one purpose (collecting money)
- ▶ To achieve another (change behavior)

UNINTENDED STRATEGIC ADJUSTMENT

Unintended consequences

$\implies$ Strategic adjustment that weakens or defeats the policy

$\implies$ Change in $a^*(p)$ that decreases U

Unintended consequences may be anticipated or unanticipated

ANTICIPATED UNINTENDED CONSEQUENCE

Anticipated unintended consequence

⇒ Strategic adjustment that weakens the policy but is anticipated and factored into the optimal design

Example: Increasing taxes

- ▶ Is meant to increase government's revenue
- ▶ But discourages work at the margin: policy makers anticipate that
- ▶ Trade-off: optimal tax trades off revenue vs distortion



Thomas Sowell

*“There are no solutions.
There are only trade-offs.”*

TRADE OFFS FROM AP

When designing institutions:

- ▶ Strong executive → decisive action but abuse risk
Strong constraints → accountability but inefficient problem solving
- ▶ Better selection → may worsen accountability
- ▶ Term limits → lower pandering but hinder accountability and selection
- ▶ Rules solve commitment problem → worsen decision making ex post
- ▶ Checks and balances prevent abuse of power → slow down responses to problems

TRADE OFFS FROM AP

When designing incentives in organizations:

- ▶ High-powered incentives → induce effort but introduce risk and incentives to cheat
- ▶ Subjective evaluations solve multitasking problem → suffer from commitment, favoritism and bias
- ▶ Making it easier to fire workers increases effort → makes the job more risky, increases compensation demands and increases pandering
- ▶ Relying on experts makes decisions more informed → may make them more biased
- ▶ Increasing number of experts leads to more informed decisions → may lead to herding and false consensus

THEORY OF THE SECOND BEST

First best: The ideal outcome you want to achieve

- ▶ In economics, usually a utilitarian optimum

The theory of the second best: Any policy to address one issue is likely to cause other distortions (Lipsey and Lancaster 1956)

Second best: a policy that *optimally* balances trade-offs

- ▶ In economics, a policy that maximizes total utility subject to frictions
- ▶ More broadly, you need to balance trade-offs in a way that takes you closest to your normative goals

UNANTICIPATED UNINTENDED CONSEQUENCE

Unanticipated unintended consequence

⇒ Strategic adjustment that weakens the policy but is unanticipated and cannot be factored into the policy

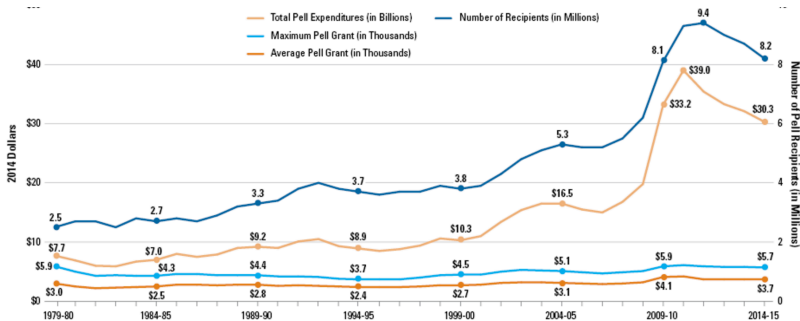
Example: CAFE and fatalities

- ▶ Higher fatalities due to people driving more trucks were most likely unanticipated

Unintended consequences can derail your policy!

EXAMPLE: FEDERAL FINANCIAL AID TO INCREASE COLLEGE AFFORDABILITY

Federal financial aid in the U.S. has been steadily growing



College affordability not so much

COST TO STUDENTS ONLY DECREASED BY 30 CENT FOR EVERY DOLLAR OF AID

Attempted goal: financial aid lowers out of pocket cost of education

Strategic adjustment: universities can decrease internal aid or raise prices and still attract students

- ▶ It is difficult for supply of college education to expand
- ▶ Colleges respond by raising prices

Increased financial aid primarily benefits universities

Caveat: This isn't true at community colleges and technical schools, where supply can expand more easily

UNINTENDED CONSEQUENCES ARE SOMETIMES HARD TO SEE EX POST

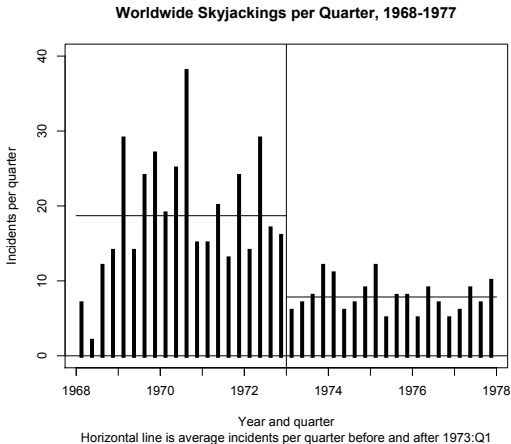
Many policies appear to work if one looks at the problem they were meant to address

But distortions arise in other areas

EXAMPLE: FIGHTING TERRORISM

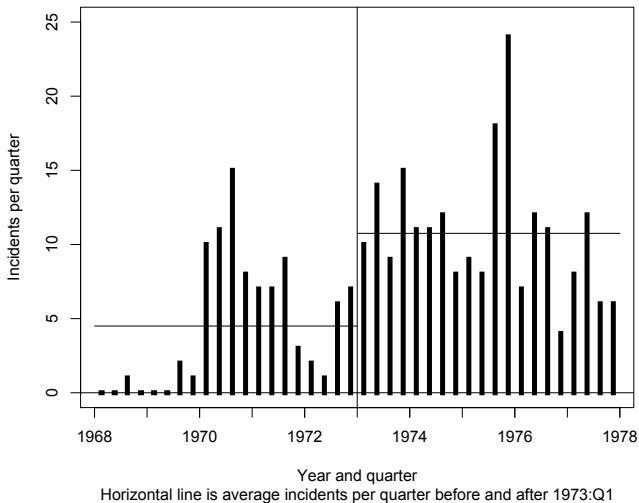
Security at airports was raised in response to skyjackings

Metal detectors were installed in 1973



SUBSTITUTION TO OTHER TACTICS

Worldwide Hostage Takings per Quarter, 1968-1977

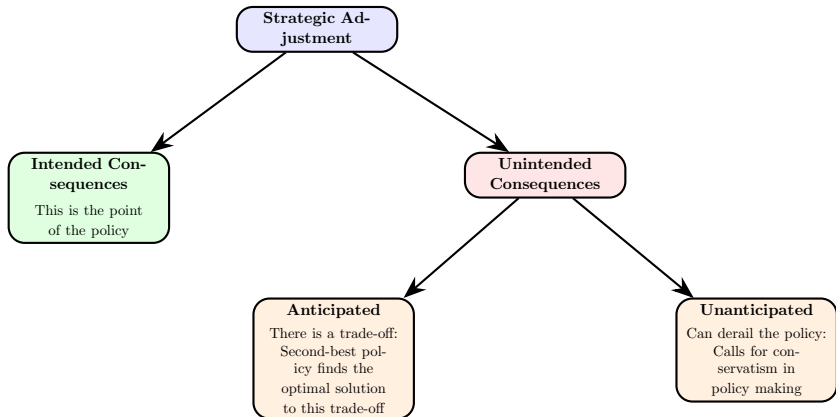


UNINTENDED BEHAVIOR IS DUE TO MISPLACED INCENTIVES

Many unintended consequences are driven by:

- ▶ **Multitasking-like problems:** incentivizing what you can measure leads to effort diversion away from what you really care about
- ▶ **Cobra effect:** incentivizing outcomes that can be achieved via undesired actions instead of incentivizing actions directly
- ▶ **Threshold effects:** attaching strong incentives to exceeding or staying below a clearly defined threshold

SUMMARY OF STRATEGIC ADJUSTMENT



TAKE AWAYS FOR POLICY MAKING

1. Remember there are always trade-offs
2. Not all inefficiencies require reforms!
3. Since there are unintended consequences of any reform, **be cautious** when deciding to undertake reforms

EXERCISE: DESIGN THE PERFECT WELFARE PROGRAM

Form groups of 3-4 people

You are policymakers

Some desirable properties:

1. Provide income support to families in genuine need
2. Be politically viable, which may include
 - ▶ Avoid discouraging able-bodied adults from working
 - ▶ Prevent fraud or abuse

Design the program. Ask yourself:

- ▶ Who must fill out forms?
- ▶ What documentation is required?
- ▶ How often must eligibility be verified?

You have 5 minutes

THE COMPLEXITY TRAP

Why did your program become **complex**?

Complexity is frequently the result of trying to

- ▶ Avoid unintended consequences
- ▶ Avoid trade offs
- ▶ Fine tune incentives
- ▶ Solve political constraints (build coalition with diverse interests)

COMPLEXITY VS. SIMPLICITY

Complex policy is expensive to administer

Complex policy is hard to navigate by those targeted by it

Complex policy creates inequalities

- ▶ Those with resources more likely to benefit
- ▶ Those with resources more likely to game the system

Complex policy facilitates corruption

- ▶ Incentives to bribe bureaucrats to avoid complexity
- ▶ Harder to detect corruption

Complexity should be an important concern for policy design

FRAUD SHOCK → DEMAND FOR “CRACKDOWN”



**During early pandemic (2020–21)
unemployment spiked**

- ▶ Pandemic unemployment insurance (UI) programs expanded quickly
- ▶ Fraud and improper payments became a **salient political issue**

Congress enacted an ID verification for pandemic UI benefits

UNINTENDED CONSEQUENCE: COMPLEXITY DETERS THE ELIGIBLE

Implementation reality:

- ▶ States layered additional antifraud controls as vulnerabilities emerged
- ▶ Some required identity verification *before* claim submission
- ▶ Remote identity proofing meant document uploads, video calls, long queues

Distributional result:

Complexity acts like a **regressive barrier**: it screens out the eligible who are least able to navigate the process

- ▶ People with limited documentation
- ▶ Limited internet or device access
- ▶ Lower administrative capacity (time, literacy, stability)

POLITICAL LOGIC

Fraud is salient $\Rightarrow$ voters demand visible action

Elected officials **pander** and tighten restrictions in visible ways

Negative downstream consequences follow much later and may be less visible

- ▶ The affected groups are frequently politically less active

TAKE AWAYS

Before reforming, ask:

1. How will people adjust strategically?
2. What might we be missing?
3. What tradeoffs are unavoidable?
4. Does complexity undermine the purpose of the policy?